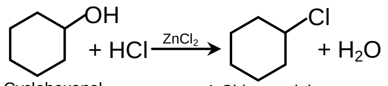
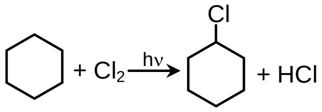
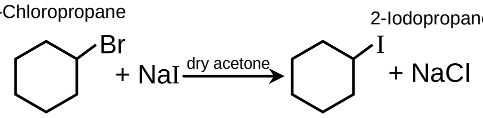
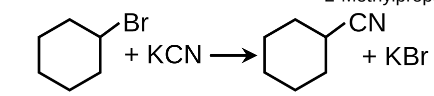
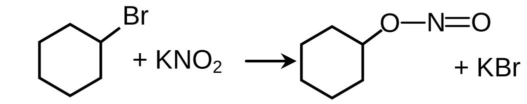
ABC-3 (Alkyl halide, Alcohol & Ether)

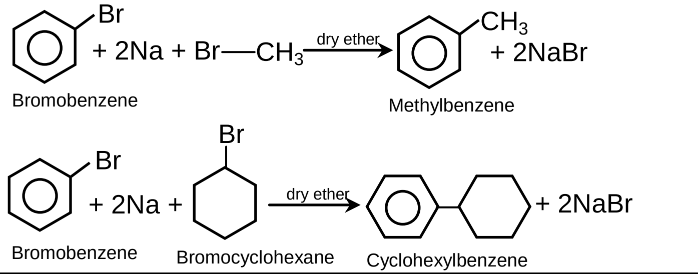
(A) ALKYL HALIDE

Preparation of alkyl halides (5-Methods)

1.	**From alcohol** (i) from ($SOCl_2$) in presence of pyridine $R{-}OH \xrightarrow[\text{heat, pyridine}]{SOCl_2} R{-}Cl + SO_2 + HCl$ It is known as Darzon method.	$\underset{\text{Ethanol}}{CH_3{-}CH_2{-}OH} \xrightarrow[\text{Pyridine}]{SOCl_2,\ \Delta} \underset{\text{Chloroethane}}{CH_3{-}CH_2{-}Cl} + \underbrace{SO_2 + HCl}_{\text{By products are gases}}$
	(ii) from PCl_5 $R{-}OH + PCl_5 \longrightarrow R{-}Cl + HCl + POCl_3$	$CH_3{-}\underset{\underset{CH_3}{\vert}}{CH}{-}OH + PCl_5 \longrightarrow CH_3{-}\underset{\underset{CH_3}{\vert}}{CH}{-}Cl + HCl + POCl_3$ Propan-2-ol 2-Chloropropane
	(iii) from PX_3 $3R{-}OH + PX_3 \xrightarrow{(PX_3 = PCl_3, PBr_3, PI_3)} 3R{-}X + H_3PO_3$	$\underset{\text{Ethanol}}{3CH_3{-}CH_2{-}OH} + PCl_3 \longrightarrow \underset{\text{Chloroethane}}{3CH_3{-}CH_2{-}Cl} + H_3PO_3$
	(iv) **Lucas test :** from HX(X= Cl, Br,I) $R{-}OH + HX \xrightarrow{ZnCl_2} R{-}X + H_2O$ (rate : 3º ROH > 2º ROH > 1º ROH)	 Cyclohexanol 1-Chlorocyclohexane
2.	**Halogenation of alkane** Halogenation take place either at high temperature (573-773 K) or in the presence of diffuse sunlight or ultraviolet light. Rate of reaction of alkanes with halogens : $F_2 > Cl_2 > Br_2 > I_2$	$\underset{\text{Methane}}{CH_4} + Cl_2 \xrightarrow{h\nu} \underset{\text{Chloromethane}}{CH_3Cl} + HCl$  Cyclohexane 1-Chlorocyclohexane
3.	**Addition of hydrogen halides** Hydrogen halides add up to alkenes to form alkyl halides. The order of reactivity of hydrogen halides is HI > HBr > HCl.	$\underset{\text{Ethene}}{CH_2{=}CH_2} + HBr \longrightarrow \underset{\text{Bromoethane}}{CH_3CH_2Br}$ $CH_3{-}\underset{\underset{CH_3}{\vert}}{C}{=}CH_2 + HCl \longrightarrow CH_3{-}\overset{\overset{Cl}{\vert}}{\underset{\underset{CH_3}{\vert}}{C}}{-}CH_3$ 2-Methylprop-1-ene 2-Chloro-2-methylpropane
4.	**Finkelstein reaction** Alkyl iodides often prepared by the reaction of alkyl chloride / bromide with NaI in dry acetone. This reaction is known as **Finkelstein** reaction.	$CH_3{-}\overset{\overset{H}{\vert}}{\underset{\underset{Cl}{\vert}}{C}}{-}CH_3 + NaI \xrightarrow{\text{Dry aceteone}} CH_3{-}\overset{\overset{H}{\vert}}{\underset{\underset{I}{\vert}}{C}}{-}CH_3 + NaCl\downarrow$ 2-Chloropropane 2-Iodopropane  1-Bromocyclohexane 1-Iodocyclohexane
5.	**Swart reaction** The synthesis of alkyl fluoride is best accomplised by heating an alkyl chloride/ bromide in the presence of a metallic fluoride such as AgF, Hg_2F_2, CoF_2 or SbF_3. The reaction is termed as **Swart** reaction.	$\underset{\text{Chloromethane}}{CH_3{-}Cl} + AgF \xrightarrow{\Delta} \underset{\text{Fluoromethane}}{CH_3{-}F} + AgCl$ $CH_3{-}CH_2{-}\overset{\overset{Cl}{\vert}}{CH}{-}CH_3 + AgF \longrightarrow$ 2-Chlorobutane $CH_3{-}CH_2{-}\underset{\underset{F}{\vert}}{CH}{-}CH_3 + AgCl$ 2-Fluorobutane

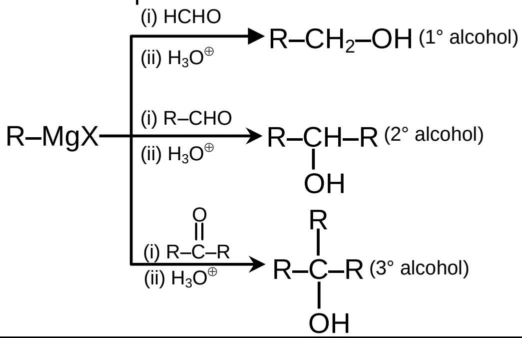
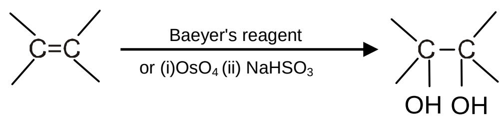
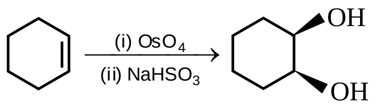
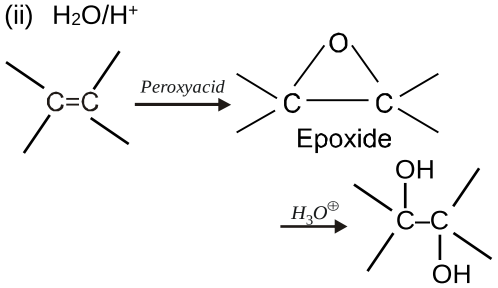
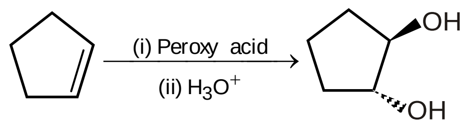
Chemical reactions of alkyl halide (4-Reactions)

1.	**Reaction of alkyl halide with** **(a) KCN** KCN is predominantly ionic and provides cyanide ions in solution. Although both carbon and nitrogen atoms are in a position to donate electron pairs, the attack takes place mainly through carbon atom and not through nitrogen atom since C–C bond is more stable than C–N bond.	$CH_3-CH(Cl)-CH_3 + KCN \longrightarrow CH_3-CH(CN)-CH_3 + KCl$ 2-Chloropropane → 2-Methylpropanenitrile  1-Bromocyclohexane Cyclohexanecarbonitrile [Only for 1º and 2º alkyl halides.] [Phenylic, vinylic and bridge head halides fail to react by this reaction.]
	(b) AgCN AgCN is mainly covalent in nature and nitrogen is free to donate electron pair forming isocyanide as the main product.	$CH_3-Cl + AgCN \longrightarrow CH_3NC + AgCl$ Methylchloride Methylisocyanide $CH_3-CH(Cl)-CH_3 + AgCN \longrightarrow CH_3-CH(NC)-CH_3 + AgCl$ 2-Chloropropane
	(c) KNO_2 Alkyl halides (R–X) react with KNO_2 to give R–O–N=O (Alkylnitrite).	$CH_3-CH(Cl)-CH_3 + KNO_2 \longrightarrow CH_3-CH(O-N=O)-CH_3 + KCl$ Iso-propylchloride Iso-Propylnitrite  Cyclohexylbromide Cyclohexylnitrite
	(d) $AgNO_2$ Alkyl halides (R–X) react with $AgNO_2$ to give $R-NO_2$ (Nitroalkane)	$CH_3-Cl + AgNO_2 \longrightarrow CH_3NO_2 + AgCl$ Chloromethane Nitromethane $CH_3-CH_2-CH_2-CH_2-Cl + AgNO_2$ 1-Clorobutane $\longrightarrow CH_3-CH_2-CH_2-CH_2-NO_2 + AgCl$ 1-Nitrobutane
	(e) Aqueous KOH Alkyl halides (R–X) when reacts with aq. KOH gives alcohol	$CH_3-CH_2-Cl + aq.\ KOH \longrightarrow CH_3-CH_2-OH + KCl$ Chloro ethane Ethanol $CH_3-CH(Cl)-CH_3 + KOH(aq.) \longrightarrow CH_3-CH(OH)-CH_3 + KCl$ 2-Chloropropane Propan-2-ol
2.	**Wurtz reaction** Alkyl halide on treatment with sodium metal in dry ether solution give higher alkanes. The reaction is known as **Wurtz** reaction. It is used for the preparation of higher alkanes containing even number of carbon atoms.	$CH_3-Br + 2Na + Br-CH_3 \xrightarrow{\text{dry ether}} CH_3-CH_3 + 2NaBr$ Bromomethane Ethane $CH_3-CH(CH_3)-Br + 2Na + Br-CH(CH_3)-CH_3$ 2-Bromopropane $\xrightarrow{\text{dry ether}} CH_3-CH(CH_3)-CH(CH_3)-CH_3 + 2NaBr$ 2,2-Dimethyl butane

3.	**Wurtz-Fittig reaction** A mixture of an alkyl halide and aryl halide give an alkyl arene when treated with sodium in dry ether.	
4.	**Williamson ether synthesis**	$CH_3Cl + CH_3O^-K^+ \longrightarrow CH_3 - O - CH_3 + KCl$ Methyl chloride, Potassium methoxide, Dimethyl ether

(B) ALCOHOL AND ETHERS

Preparation of alcohols (4-Methods)

1.	**From Grignard reagents [R–Mg–X]** Grignard Reagent is an organometallic compound. Grignard Reagent acts as base and nucleophile. 	(a) $CH_3MgBr + HCHO \xrightarrow{Ether} \xrightarrow{H_3O^+} CH_3–CH_2–OH$ (b) $PhMgBr + CH_3–CHO \xrightarrow{Ether} \xrightarrow{H_3O^+} CH_3–CH(Ph)–OH$ (c) $CH_3–CH_2MgBr + CH_3–C(=O)–CH_3 \xrightarrow{Ether} \xrightarrow{H_3O^+} CH_3–C(CH_3)(CH_2CH_3)–OH$
2.	**Syn hydroxylation of alkenes** **Reagents :** **a.** Baeyer's reagent: [cold dilute 1% alkaline $KMnO_4$] or **b.** (i) OsO_4 (ii) $NaHSO_3$  **Remark** It is syn addition. Both OH groups add on the same side of pi-bond.	(a) $H_2C = CH_2 \xrightarrow[(ii)\ NaHSO_3]{(i)\ OsO_4} H_2C(OH)–CH_2(OH)$ (b) $CH_3–CH=CH_2 \xrightarrow{1\%\ alk.\ KMnO_4} CH_3–CH(OH)–CH_2(OH)$ (c) 
3.	**Anti hydroxylation of alkene** **Reagent**: (i) Peroxyacid (RCOOOH) (ii) H_2O/H^+ 	(a) $CH_3—CH = CH_2 \xrightarrow[H_3O^+]{PhCO_3H} CH_3—CH(OH) — CH_2(OH)$ (b) $CH_3–CH=CH–CH_3 \xrightarrow[H_3O^+]{PAA} CH_3–CH(OH)–CH(OH)–CH_3$ (c) 

	Remark It is anti addition. Both OH groups add on the opposite side of pi-bond. Peroxyacid may be any one of these: **a.** m-CPBA (Metachloro perbenzoic acid) **b.** PAA (Peracetic acid) **c.** PBA (Per benzoic acid) **d.** TFPAA (Trifluoro peracetic acid).	
4.	**Acid catalyzed hydration of alkene** **Reagents: dil.** H_2SO_4 or H_2O/H^+ or H_3O^+ $>C=C< \xrightarrow{H_2O/H^{\oplus}} -\underset{\underset{H}{\vert}}{\overset{\vert}{C}}-\underset{\underset{OH}{\vert}}{\overset{\vert}{C}}-$ **Remarks:** Markovnikov's addition	(a) $CH_3-CH=CH_2 + H_2O \xrightarrow{H^{\oplus}} CH_3-\underset{\underset{OH}{\vert}}{CH}-CH_3$ (b) $CH_3-CH=CH-CH_3 \xrightarrow{H_2O/H^{\oplus}} CH_3-\underset{\underset{H}{\vert}}{CH}-\underset{\underset{OH}{\vert}}{CH}-CH_3$

Preparation of ethers (2-Methods)

1.	**Williamson synthesis** It is an important laboratory method for the preparation of symmetrical and unsymmetrical ethers. In this method, an alkyl halide is allowed to react with sodium alkoxide. $RX + R'\overset{-}{O}-\overset{+}{Na} \longrightarrow R'-O-R + NaX$	(a) $CH_3-X + CH_3-CH_2-CH_2O-Na \longrightarrow CH_3-O-CH_2CH_2CH_3$ (b) $CH_3CH_2-X + PhO-Na \longrightarrow Ph-O-CH_2-CH_3$ (c) $Ph-CH_2-X + Ph-CH_2-O-Na \longrightarrow PhCH_2-O-CH_2-Ph$
2.	**From alcohol** Alcohols undergo dehydration in the presence of protic acids (H_2SO_4, H_3PO_4) at 413 K temperature (140ºC). $R-OH \xrightarrow[413\,K]{H_2SO_4} R-O-R$ **Remark** : Alcohols undergo dehydration by heating with conc. H_2SO_4 at 443K and give alkene.	(a) $CH_3-CH_2-OH \xrightarrow[413\,K]{H_2SO_4} C_2H_5-O-C_2H_5$ (b) $CH_3-OH + CH_3OH \xrightarrow[413\,K]{H_2SO_4} CH_3-O-CH_3$ (c) $CH_3-OH + CH_3-CH_2-OH \xrightarrow[413\,K]{H_2SO_4} CH_3-O-CH_3$ $+ CH_3-CH_2-O-CH_2-CH_3 + CH_3-O-CH_2-CH_3$

Chemical reactions of alcohols (5-Reactions)

1.	**Reaction with HX:** $R-OH + HX \longrightarrow R-X + H_2O$ The reactions of primary and secondary alcohols with HCl require a catalyst ($ZnCl_2$). With tertiary alcohols, the reaction conduct by simply shaking with concentrated HCl at room temperature.	(a) $CH_3-CH_2-OH \xrightarrow{HBr} CH_3-CH_2-Br$ (b) $CH_3-\underset{\underset{CH_3}{\vert}}{CH}-OH \xrightarrow[ZnCl_2]{HCl} CH_3-\underset{\underset{CH_3}{\vert}}{CH}-Cl$
2.	**Reaction with phosphorus trihalides:** $3R-OH + PX_3 \longrightarrow 3R-X + H_3PO_3$ $PX_3 = PCl_3, PBr_3, PI_3$	(a) $CH_3-CH_2-CH_2-OH \xrightarrow{PBr_3} CH_3-CH_2-CH_2-Br$ (b) $CH_3-CH_2-\overset{\overset{CH_3}{\vert}}{CH}-CH_2-OH \xrightarrow{PCl_3}$ $CH_3-CH_2-\overset{\overset{CH_3}{\vert}}{CH}-CH_2-Cl$

3.	**Reaction with PCl_5** $R{-}OH + PCl_5 \longrightarrow R{-}Cl + HCl + POCl_3$	(a) $PhCH_2{-}OH \xrightarrow{PCl_5} PhCH_2Cl + POCl_3$ (b) $CH_3{-}\underset{\substack{\vert\\CH_3}}{CH}{-}OH \xrightarrow{PBr_5} CH_3{-}\underset{\substack{\vert\\CH_3}}{CH}{-}Br$
4.	**Reaction with thionyl chloride in presence of pyridine:** $R{-}OH + SOCl_2 \xrightarrow{pyridine} R{-}Cl + SO_2 + HCl$	(a) $CH_3{-}CH_2{-}OH \xrightarrow[pyridine]{SOCl_2} CH_3{-}CH_2{-}Cl$ (b) $Ph{-}CH_2{-}OH \xrightarrow[pyridine]{SOBr_2} Ph{-}CH_2{-}Br$
5.	**Reaction with conc. H_2SO_4 / Δ** $CH_3{-}\overset{\substack{H\\\vert}}{\underset{\substack{\vert\\H}}{C}}{-}\overset{\substack{H\\\vert}}{\underset{\substack{\vert\\OH}}{C}}{-}CH_3 \xrightarrow[\Delta]{Conc.\ H_2SO_4}$ $CH_3{-}CH{=}CH{-}CH_3$ **Remarks :** **i.** Ease of dehydration of alcohol $\Rightarrow 3^o > 2^o > 1^o$ **ii.** More stable alkene **(Saytzeff Alkene)** is formed as major product	(a) $CH_3{-}\underset{\substack{\vert\\OH}}{CH}{-}CH_3 \xrightarrow[\Delta]{Conc.\ H_2SO_4} CH_3{-}CH{=}CH_2$ (b) $CH_3{-}\overset{\substack{CH_3\\\vert}}{\underset{\substack{\vert\\OH}}{C}}{-}CH_3 \xrightarrow[\Delta]{Conc.\ H_2SO_4} CH_3{-}\overset{\substack{CH_3\\\vert}}{C}{=}CH_2$ (c) $CH_3{-}CH_2{-}\overset{\substack{CH_3\\\vert}}{\underset{\substack{\vert\\OH}}{C}}{-}CH_3 \xrightarrow[\Delta]{Conc.\ H_2SO_4} CH_3{-}CH{=}\overset{\substack{CH_3\\\vert}}{C}{-}CH_3$

$CH_3{-}CH_2{-}OH$ (1° alcohol) $\xrightarrow{Conc.\ H_2SO_4}$

- $\xrightarrow{80^oC}$ $CH_3{-}CH_2{-}OSO_3H$ [Alkyl hydrogen sulphate]
- $\xrightarrow{140^oC}$ $CH_3{-}CH_2{-}O{-}CH_2{-}CH_3$ [ether]
- $\xrightarrow{170^oC}$ $CH_2{=}CH_2$ [alkene]

* 2° and 3° alcohols generally give alkene at little less temperature than required for 1° alcohols.

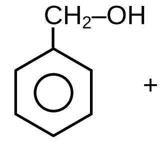
Test of alcohols (3-Test)

S.No	Test / Reagent	Observation
1	**Cerric ammonium nitrate test** **Reagent :** $[(NH_4)_2Ce(NO_3)_6]$	Red colour compound is formed
2	**Lucas reagent** **Reagent :** [Conc. HCl + anhyd. $ZnCl_2$] **Remarks :** It gives white turbidity or cloudiness with alcohols (OH groups attached with sp^3 hybridised carbon).	(a) **1° alcohol** does not give appreciable reaction. White turbidity is obtained on heating in 30 minutes. (b) **2° alcohol** gives white turbidity in 5 minutes. (c) **3° alcohol** gives white turbidity immediately.
3	**Victor Mayer test** **Reagent :** (i) $P + I_2$ (ii) $AgNO_2$ (iii) HNO_2 (iv) Base	(a) **1° alcohol –** Blood red colour. (b) **2° alcohol –** Blue colour. (c) **3° alcohol –** No colour.

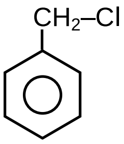
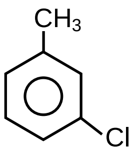
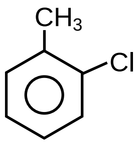
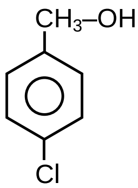
Exercise

ONLY ONE OPTION CORRECT TYPE

PART-A (Alkyl halides)

1.  $+ PCl_3 \longrightarrow [X] + H_3PO_3$

Product [X] is :

(A) 
(B) 
(C) 
(D) 

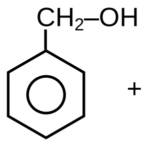
2. $CH_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-OH + HBr \longrightarrow [Y] + H_2O$

Product [Y] is :

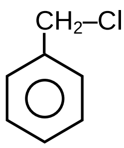
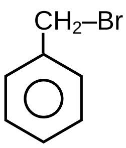
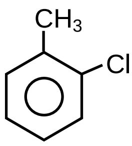
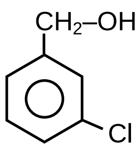
(A) $CH_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-Br$ (B) $CH_3-\underset{CH_2-Br}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-H$ (C) $CH_3-CH_2-CH_2-Br$ (D) $CH_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-CH_2-Br$

3. $[X] + PCl_5 \longrightarrow CH_3-CH_2-CH_2-Cl + HCl + POCl_3$
Reactant [X] is -

(A) $CH_3-\overset{CH_3}{\overset{|}{C}H}-OH$ (B) $CH_3-CH_2-CH_2-OH$ (C) $CH_3-CH_2-CH_3$ (D) $CH_3-\overset{CH_3}{\overset{|}{C}H}-CH_3$

4.  $+ PCl_5 \longrightarrow [X] + HCl + POCl_3$

Product [X] is :

(A) 
(B) 
(C) 
(D) 

5. $CH_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-CH_3 + Cl_2 \xrightarrow{h\nu} [X] + HCl$

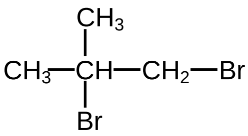
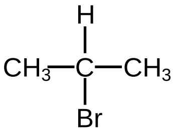
Product [X] is -

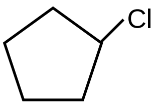
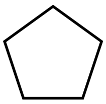
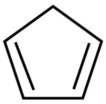
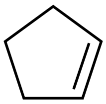
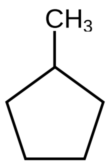
(A) $CH_3-\underset{Cl}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-CH_3$ (B) $CH_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-CH_2-Cl$ (C) $CH_3-\overset{CH_3}{\overset{|}{C}H}-CH_2-Cl$ (D) $CH_3-\underset{C_2H_5}{\overset{C_2H_5}{\underset{|}{\overset{|}{C}}}}-Cl$

6. Iso-butane $\xrightarrow{Cl_2/h\nu}$
No. of monochloro structural isomeric product
(A) 1 (B) 2 (C) 3 (4) 4

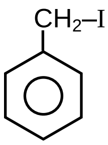
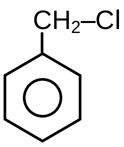
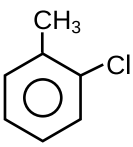
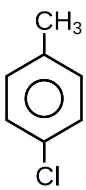
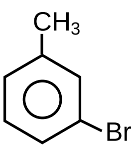
7. Iso octane $\xrightarrow{Cl_2/h\nu}$
No. of monochloro structural isomeric product
(A) 7 (B) 2 (C) 3 (D) 4

8. Cyclopentane $\xrightarrow{Cl_2/h\nu}$
No. of monochloro structural isomeric product
(A) 1 (B) 2 (C) 3 (D) 4

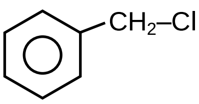
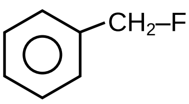
9. $CH_3–CH=CH_2 + HBr \longrightarrow [Y]$; Major product [Y] is :
(A) 
(B) $CH_3—CH_2—CH_2—Br$
(C) 
(D) $CH_3—CH_2—CH_3$

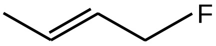
10. [Z] + HCl ⟶ 
Reactant [Z] is :
(A) 
(B) 
(C) 
(D) 

11. $CH_3–CH_2–Cl + NaI \xrightarrow{\text{Dry acetone}} [X] + NaCl$
Product [X] is -
(A) $CH_3–CH_3$ (B) $CH_3–CH_2–I$ (C) $CH_3–I$ (D) $CH_3–CH_2–CH_2–CH_3$

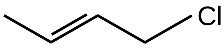
12. [Z] + NaI $\xrightarrow{\text{Dry acetone}}$ 
Reactant Z is :
(A) 
(B) 
(C) 
(D) 

13. $CH_3–CH_2–Cl + AgF \longrightarrow [X] + AgCl$
Product [X] is -
(A) $CH_3–CH_3$ (B) $CH_3–CH_2–F$ (C) $CH_3–CHF_2$ (D) $CH_3–F$

14.  + [Y] ⟶ 
Reagent [Y] can be :
(A) AgF (B) Hg_2F_2 (C) SbF_3 (D) All

15. [Z] + CoF_2 $\longrightarrow$ 

Reactant [Z] is :

(A)  (B)  (C)  (D) 

16. $CH_3-CH_2-CH_2-Cl + KNO_2 \longrightarrow [X] + KCl$

Product [X] is :

(A) $CH_3-CH_2-CH_2-O-N=O$ (B) $CH_3-CH_2-CH_2-NO_2$

(C) $CH_3-CH_2-CH_3$ (D) $CH_3-CH=CH_2$

17. $CH_3-CH_2-CH_2-Cl + AgNO_2 \longrightarrow [X] + AgCl$

Product [X] is :

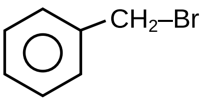
(A) $CH_3-CH_2-CH_2-O-N=O$ (B) $CH_3-CH_2-CH_2-NO_2$

(C) $CH_3-CH_2-CH_3$ (D) $CH_3-CH=CH_2$

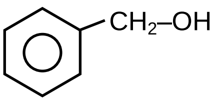
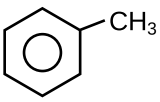
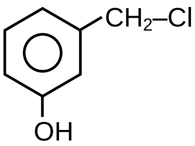
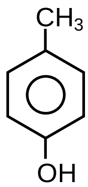
18. $CH_3-\overset{\displaystyle CH_3}{\underset{\displaystyle CH_3}{C}}-O-K^+ + CH_3-Br \longrightarrow [Y] + KBr$

Product [Y] is :

(A) $CH_3-\overset{\displaystyle CH_2}{\overset{\|}{C}}-CH_3$ (B) $CH_3-\overset{\displaystyle CH_3}{C}-CH_2-CH_3$ (C) $CH_3-\overset{\displaystyle CH_3}{\underset{\displaystyle CH_3}{C}}-O-CH_3$ (D) CH_3-CH_3

19.  + KOH (aq.) $\longrightarrow$ [X] + KBr

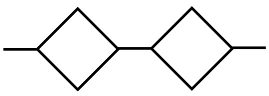
Product [X] is :

(A)  (B)  (C)  (D) 

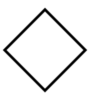
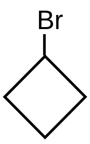
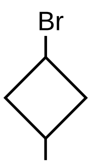
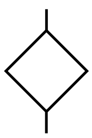
20. In wurtz reaction, $CH_3-\overset{\displaystyle CH_3}{CH}-\overset{\displaystyle CH_3}{CH}-CH_3$ can be prepared from which of the following compound?

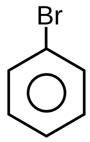
(A) $CH_3-\overset{\displaystyle CH_3}{\underset{\displaystyle H}{C}}-Br$ (B) $CH_3-CH_2-CH_2-Br$

(C) CH_3-CH_2-Br (D) $CH_3-\overset{\displaystyle CH_3}{\underset{\displaystyle CH_3}{C}}-CH_2Br$

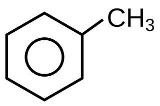
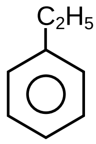
21. [X] + Na $\xrightarrow{\text{Dry ether}}$ 

Reactant [X] is -

(A)  (B)  (C)  (D) 

22.  + C_2H_5Br + 2Na $\xrightarrow{\text{Dry ether}}$ [X] + 2NaBr

Major product [X] is -

(A)  (B)  (C) $CH_3–CH_2–CH_2–CH_3$ (D) None of these

PART-B (Alcohols & Ethers)

23. Which carbonyl compound is used for following interconversion:

Carbonyl compound + $CH_3–CH_2–MgBr$ + X $\xrightarrow{H_3O^{\oplus}}$ $CH_3–CH_2–\underset{\displaystyle OH}{\overset{\displaystyle CH_3}{C}}–CH_3$

(A) $CH_3–CHO$ (B) HCHO (C) $CH_3–\underset{\displaystyle \| \atop O}{C}–CH_3$ (D) $CH_3–CH_2–CHO$

24. Which is correct product for given reaction?

$CH_3–CH_2–CHO$ + PhMgBr $\xrightarrow{H_3O^{\oplus}}$ P

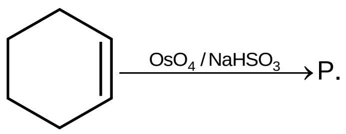
(A) $CH_3–CH_2–\underset{\displaystyle Ph}{CH}–OH$ (B) $CH_3–\underset{\displaystyle Ph}{\overset{\displaystyle CH_3}{C}}–OH$

(C) $CH_3–\overset{\displaystyle CH_3}{CH}–\underset{\displaystyle Ph}{CH_2}–OH$ (D) $Ph–CH_2–CH_2–CH_2–OH$

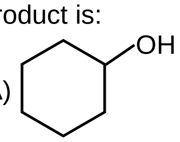
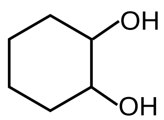
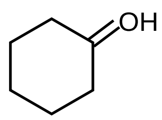
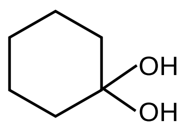
25. Which of the following carbonyl compound will give 1° alcohol after reaction with Grignard reagent followed by acidification:

(A) HCHO (B) $CH_3–CHO$ (C) $CH_3–CH_2–CHO$ (D) $CH_3–\underset{\displaystyle \| \atop O}{C}–CH_2–CH_3$

26. For the following reaction:



Product is:

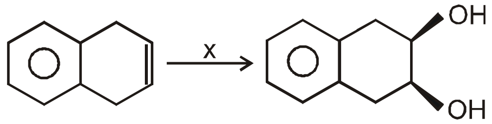
(A)  (B)  (C)  (D) 

27. For the following reaction:

$CH_3–CH=CH–CH_3 \longrightarrow CH_3–\underset{\displaystyle OH}{CH}–\underset{\displaystyle OH}{CH}–CH_3$

Reagent is :

(A) HCl (B) Dil. H_2SO_4 (C) Bayer's reagent (D) H_2/Ni

28. 

Reagent x will be :

(A) 1% alkaline $KMnO_4$ (Baeyer's reagent) (B) OsO_4 /$NaHSO_3$

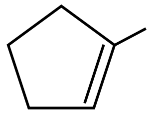
(C) Peracid/H_3O^+ (D) A and B both

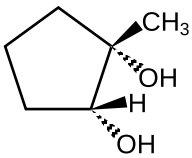
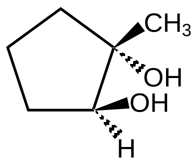
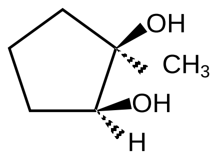
29. $CH_3–CH=CH–CH_3 \xrightarrow[\text{(ii)} H_3O^{\oplus}]{\text{(i) peroxy acid}} P$

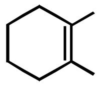
Product is:

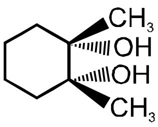
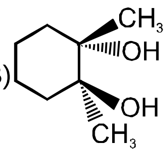
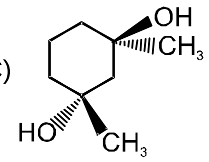
(A) $CH_3–CH_2–CH(OH)–CH_2(OH)$ (B) $CH_3–CH(OH)–CH(OH)–CH_3$

(C) $CH_3–CH_2–CH_2–CH(OH)_2$ (D) $CH_2(OH)–CH_2–CH_2–CH_2(OH)$

30.  $\xrightarrow{CH_3COOH / H_3O^{\oplus}}$ Product ; Product will be:

(A)  (B)  (C)  (D) None of these

31.  $\xrightarrow[(2) H_3O^+]{(1) \text{Peroxyacid}}$ P ; Product is :

(A)  (B)  (C)  (D) None of these

32. What is the final product of the following reaction?

$Ph–OH \xrightarrow{NaOH} \xrightarrow{CH_3–Cl}$

(A) $Ph–O–CH_3$ (B) $Ph–CH_3$ (C) $Ph–O–CH_2–CH_3$ (D) $Ph–Cl$

33. In the following reaction:

$C_2H_5ONa + C_2H_5Cl \longrightarrow$ product

Product is:

(A) $C_2H_5–O–CH_3$ (B) $C_2H_5–O–C_2H_5$ (C) $CH_3–O–CH_3$ (D) $C_2H_5–C_2H_5$

34. Predict the suitable reagent for the following conversion:

$$CH_3–CH_2–OH \longrightarrow CH_3–CH_2–O–CH_2–CH_3$$

(A) dil. H_2SO_4 (B) conc. H_2SO_4 / 200°C

(C) conc. H_2SO_4 / 140°C (D) Al_2O_3

35. In the following reaction:

$CH_3–OH + CH_3–CH_2–OH \xrightarrow[100–140°C]{Conc. H_2SO_4}$ product:

Which is not a possible product?

(A) $CH_3–O–CH_2–CH_3$ (B) $CH_3–O–CH_3$

(C) $CH_3–O–CH_2–CH_2–CH_3$ (D) $CH_3–CH_2–O–CH_2–CH_3$

36. Predict the reagent for the following reaction:

$$Ph-CH_2-OH \longrightarrow Ph-CH_2-O-CH_2-Ph$$

(A) dil. H_2SO_4 (B) $KMnO_4$ (C) $LiAlH_4$ (D) conc. H_2SO_4 / 140°C

37. Which alcohol gives instant turbidity with Lucas reagent?

(A) CH_3-CH_2-OH (B) $CH_3-\underset{OH}{\underset{|}{CH}}-CH_3$ (C) $CH_3-\overset{CH_3}{\overset{|}{\underset{OH}{\underset{|}{C}}}}-CH_3$ (D) $CH_3-CH_2-CH_2-OH$

38. Which alcohol give white turbidity in 5 minutes with Lucas reagent?

(A) CH_3-CH_2-OH (B) $Ph-\overset{Ph}{\overset{|}{\underset{OH}{\underset{|}{C}}}}-CH_3$

(C) $CH_3-\underset{OH}{\underset{|}{CH}}-CH_2-CH_3$ (D) $CH_3-\overset{CH_3}{\overset{|}{\underset{OH}{\underset{|}{C}}}}-CH_3$

39. Which test is used to distinguish 1°, 2°, and 3° alcohols?

(A) Victor Mayer test (B) Iodoform test

(C) $NaHCO_3$ test (D) Bayer's test

40. Which test is not given by alcohols?

(A) Lucas Test (B) Neutral $FeCl_3$ Test

(C) Victor major test (D) Cerric ammonium nitrate test

Answers

1.	(A)	**2.**	(A)	**3.**	(B)	**4.**	(A)	**5.**	(B)
6.	(B)	**7.**	(D)	**8.**	(A)	**9.**	(C)	**10.**	(C)
11.	(B)	**12.**	(A)	**13.**	(B)	**14.**	(D)	**15.**	(C)
16.	(A)	**17.**	(B)	**18.**	(C)	**19.**	(A)	**20.**	(A)
21.	(C)	**22.**	(B)	**23.**	(C)	**24.**	(A)	**25.**	(A)
26.	(B)	**27.**	(C)	**28.**	(D)	**29.**	(B)	**30.**	(B)
31.	(B)	**32.**	(A)	**33.**	(B)	**34.**	(C)	**35.**	(C)
36.	(D)	**37.**	(C)	**38.**	(C)	**39.**	(A)	**40.**	(B)